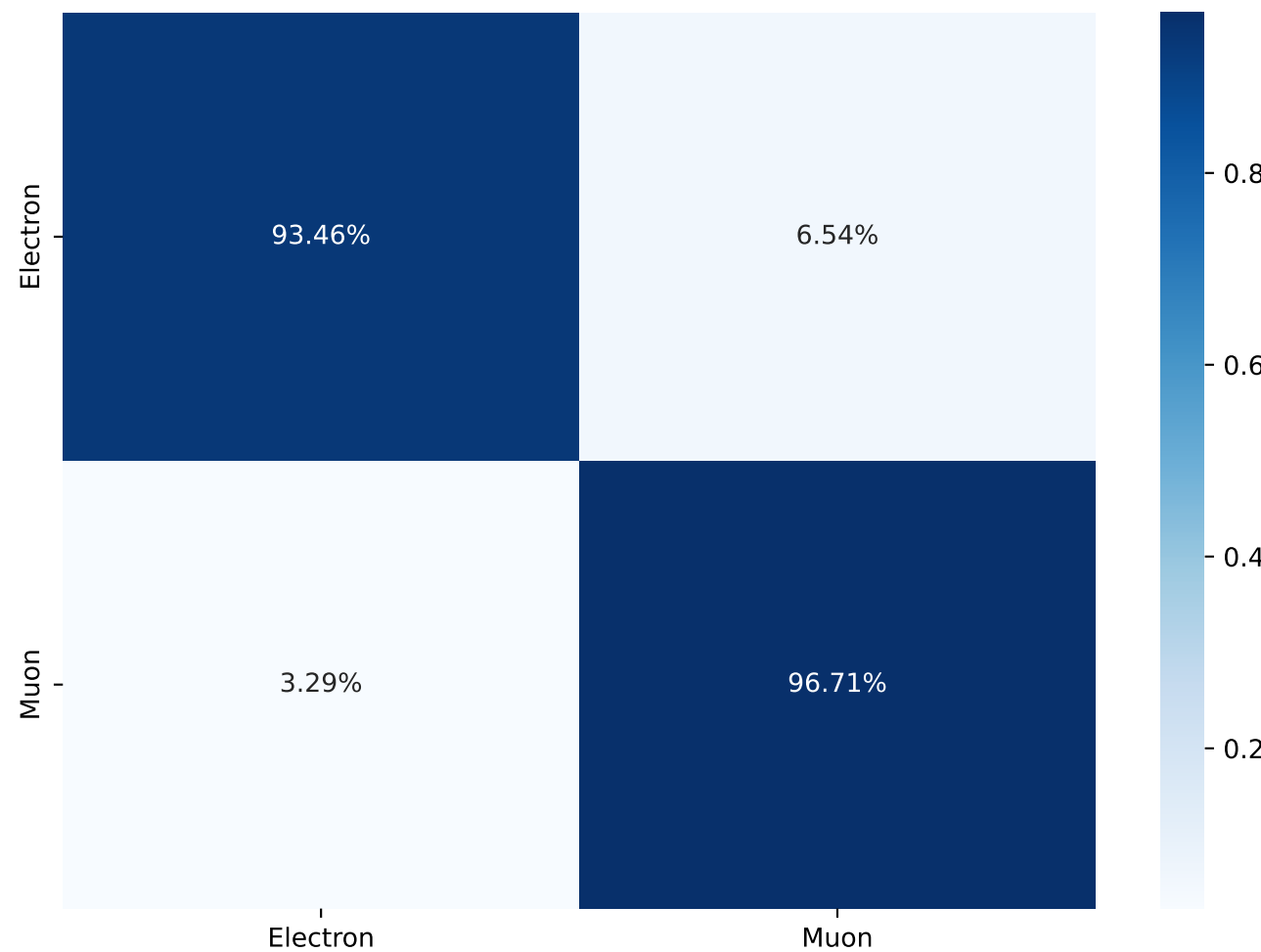
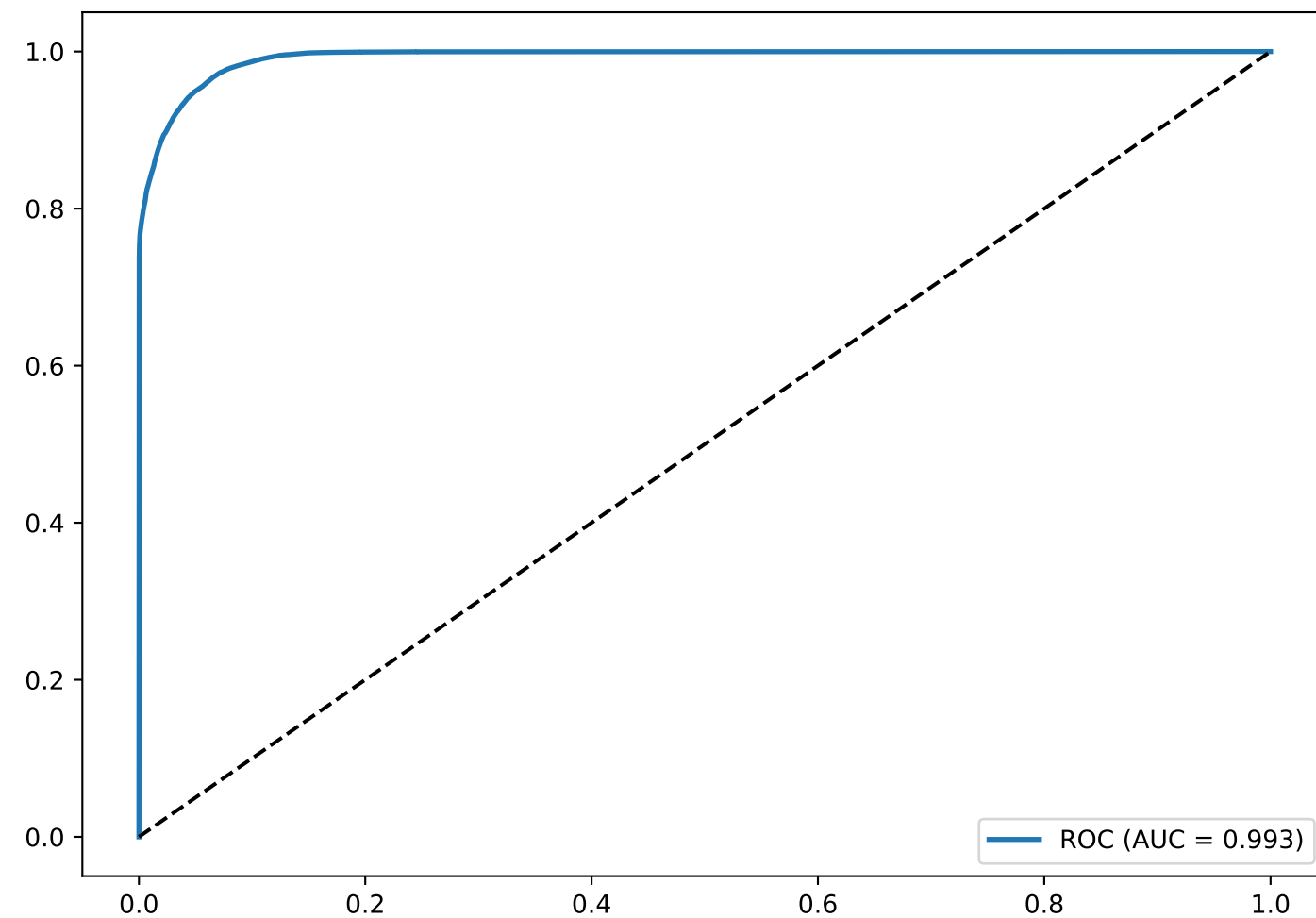


EtaRange = FullRange (TEST DATA) - Accuracy: 0.962

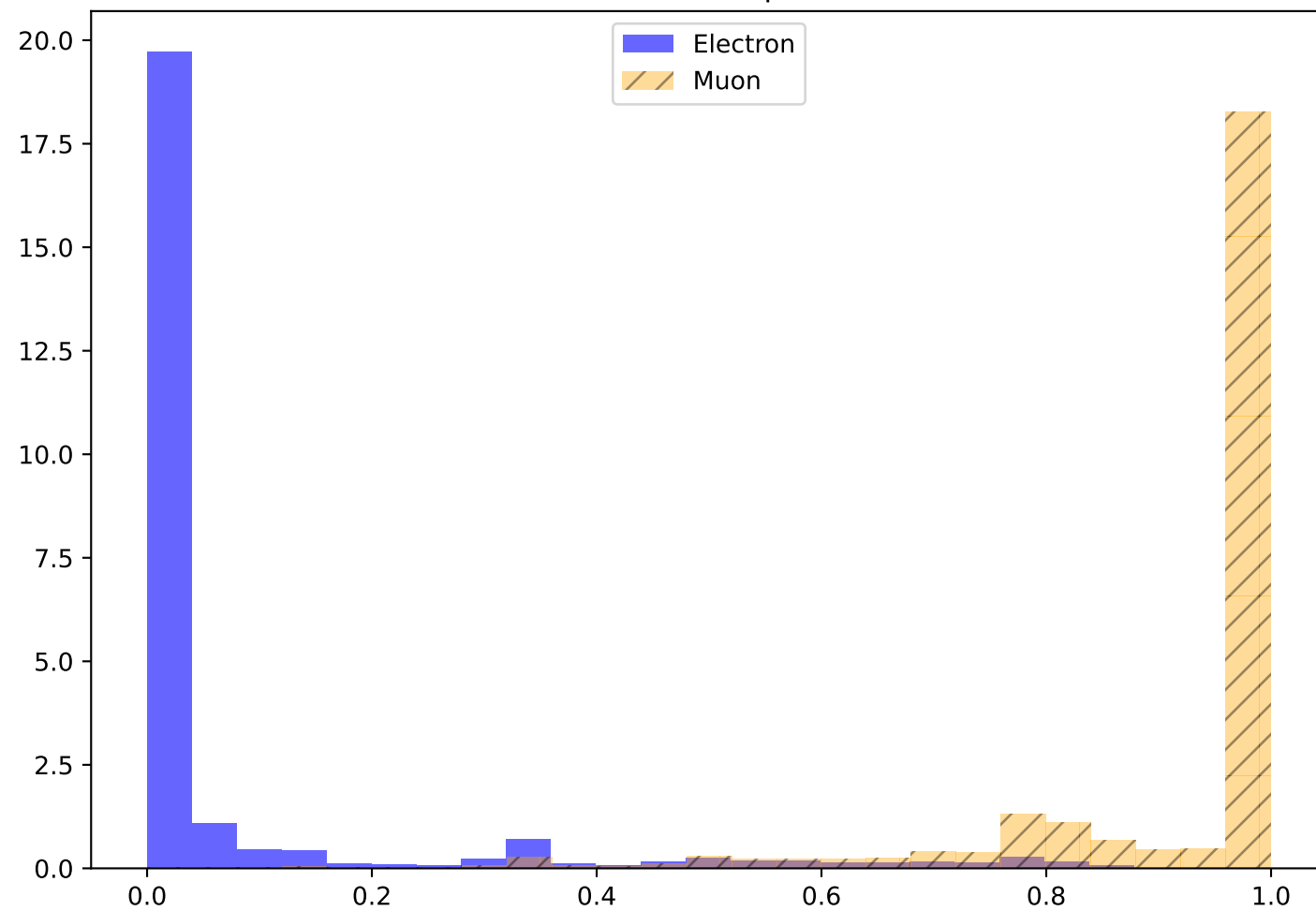
Confusion Matrix



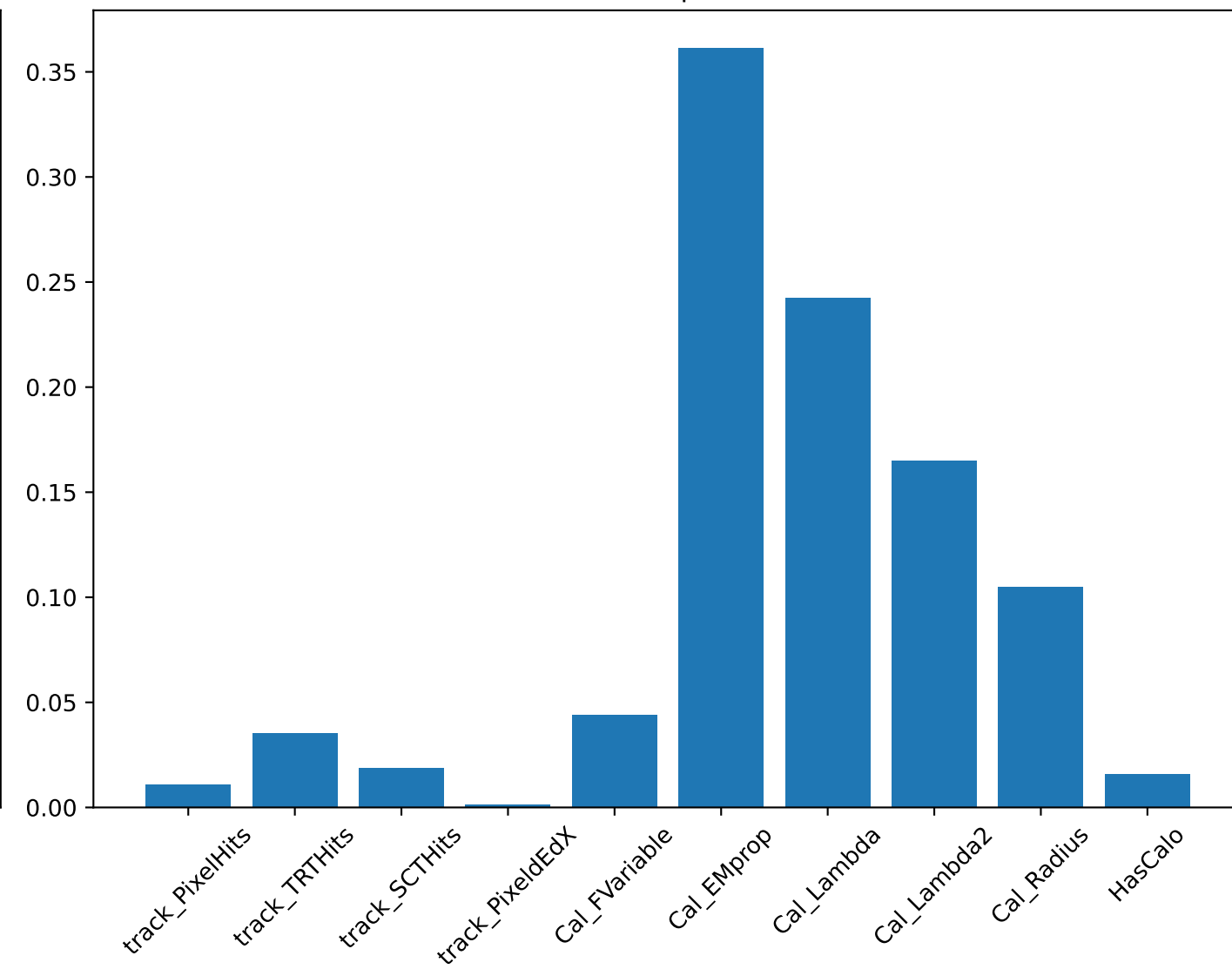
ROC Curve



Classifier Response

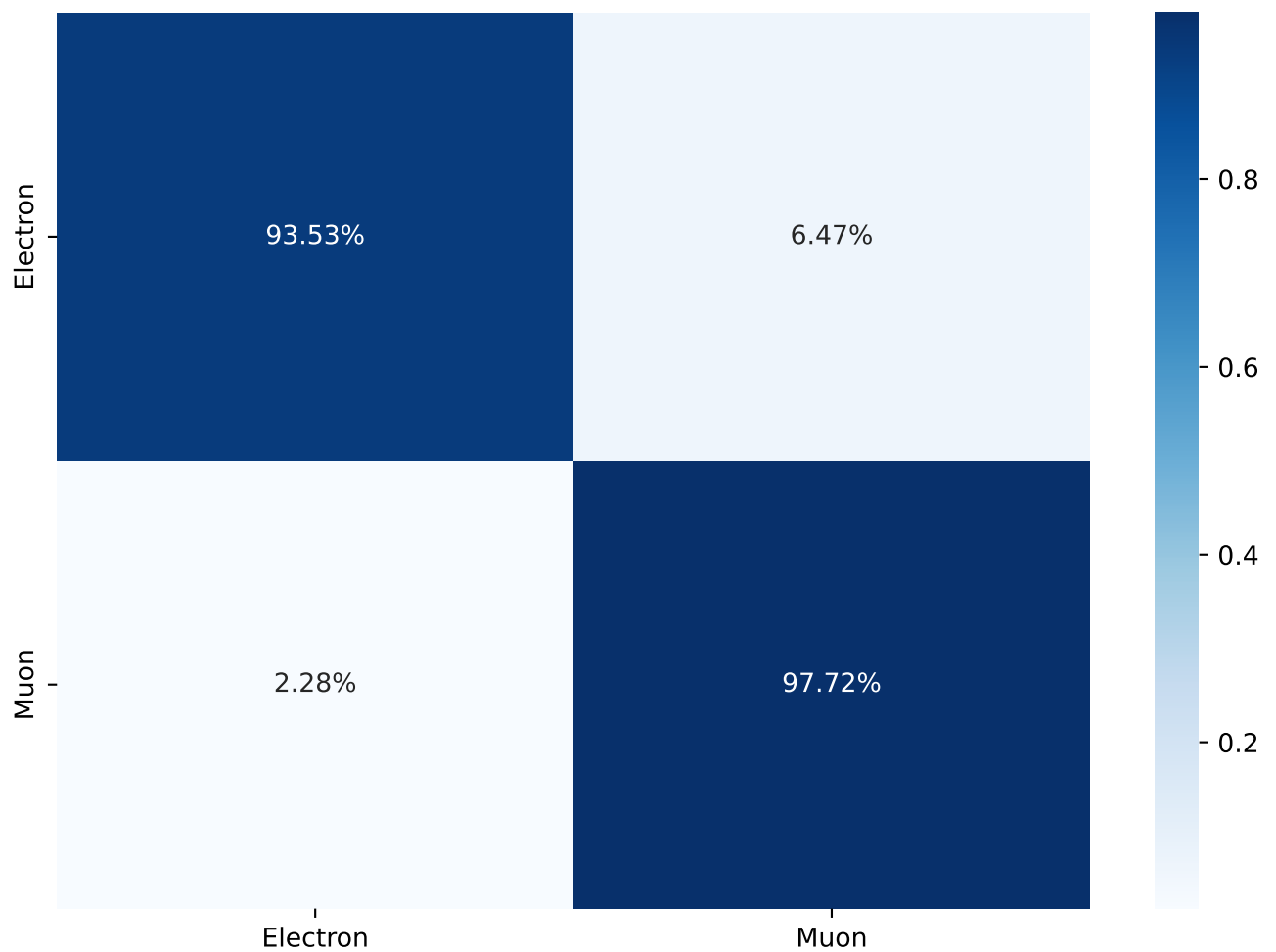


Feature Importance

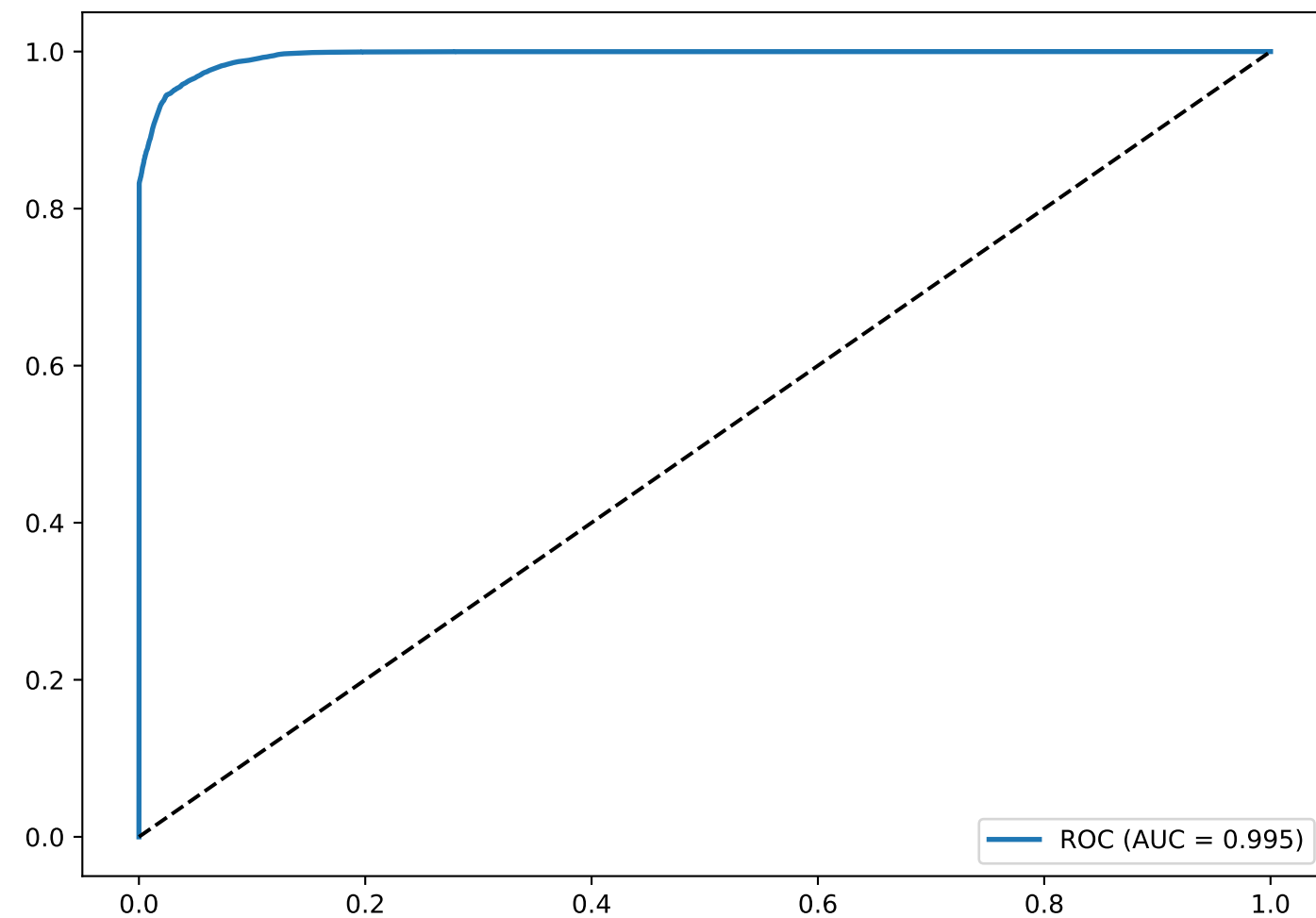


EtaRange = <1.0 (TEST DATA) - Accuracy: 0.969

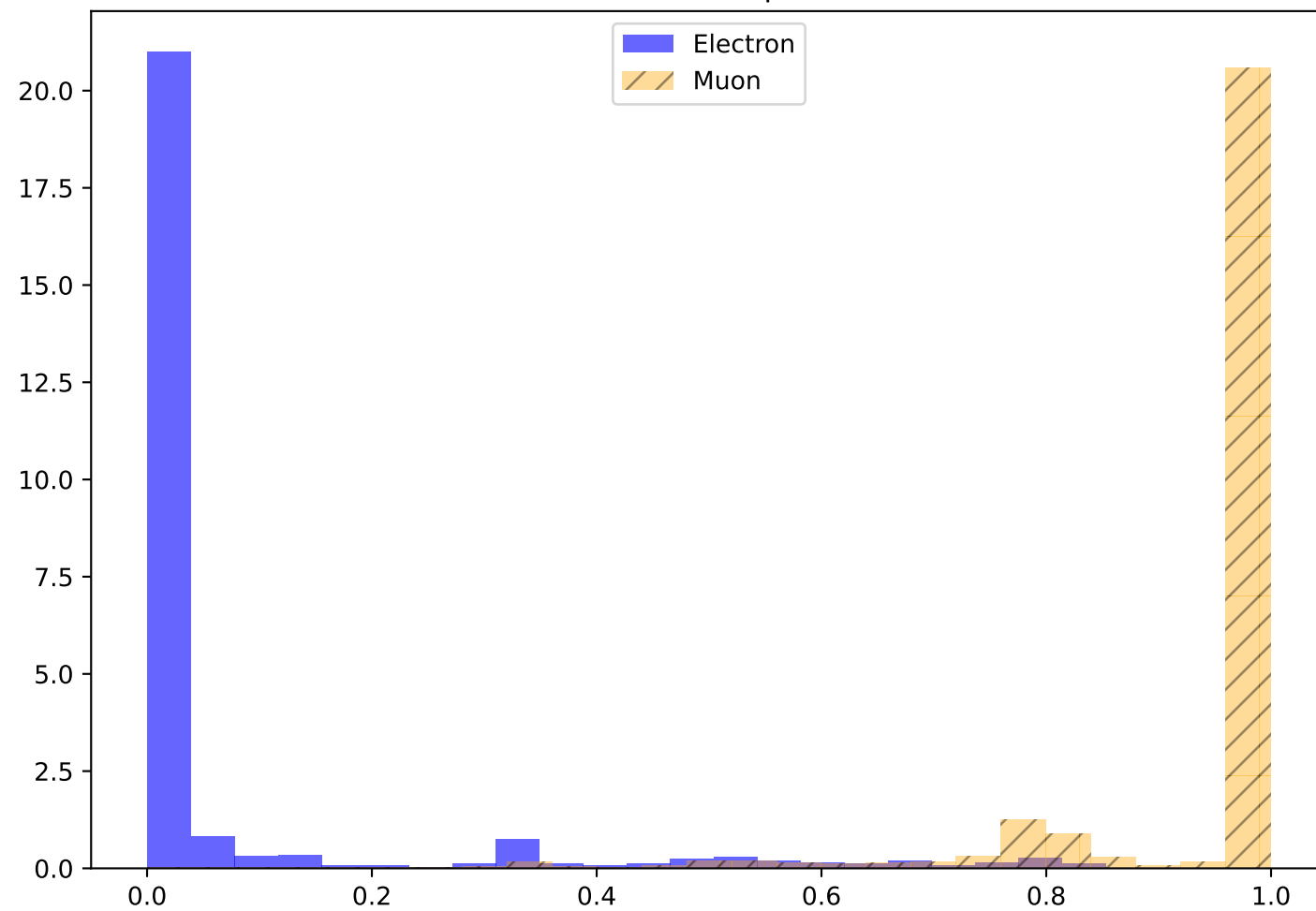
Confusion Matrix



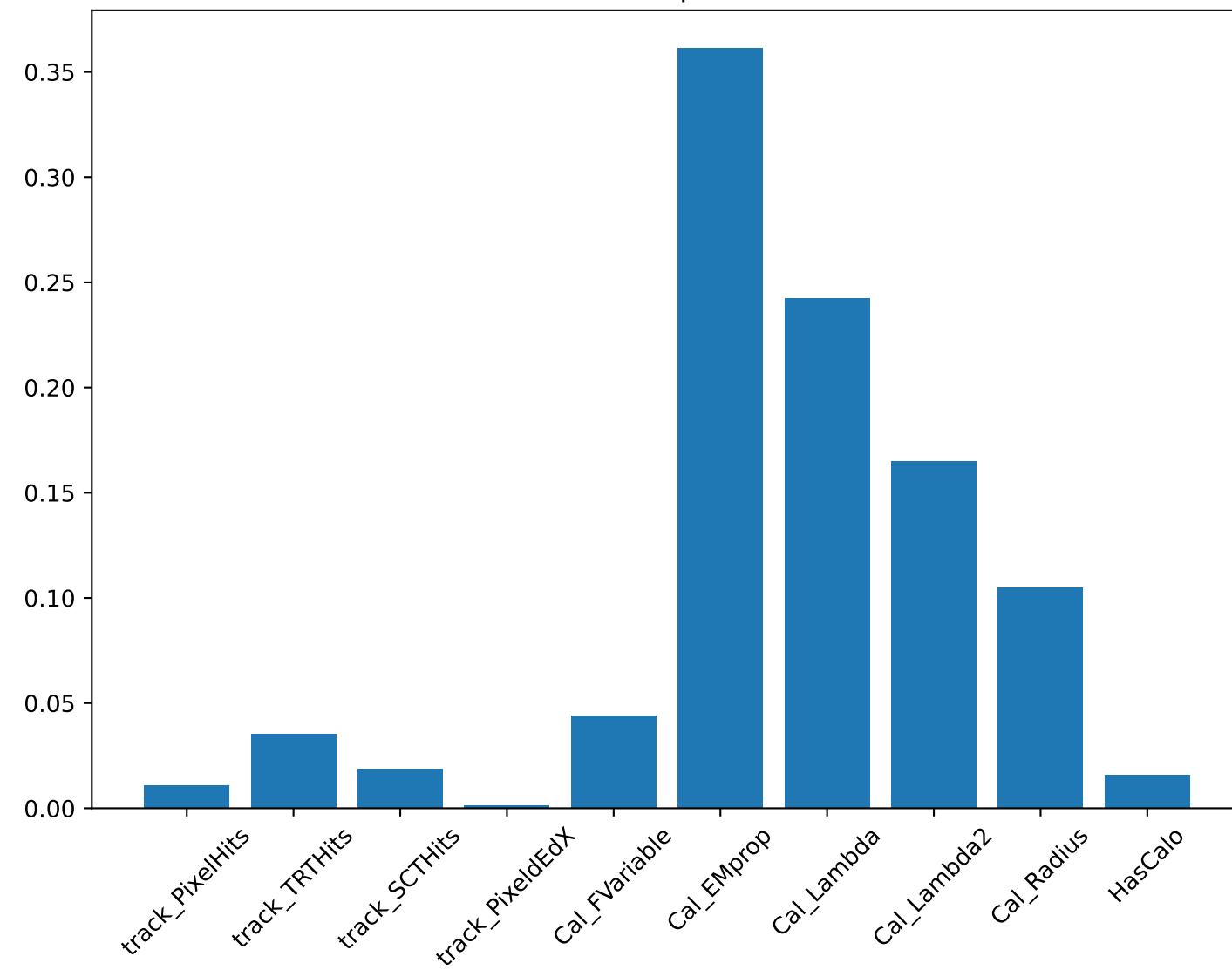
ROC Curve



Classifier Response

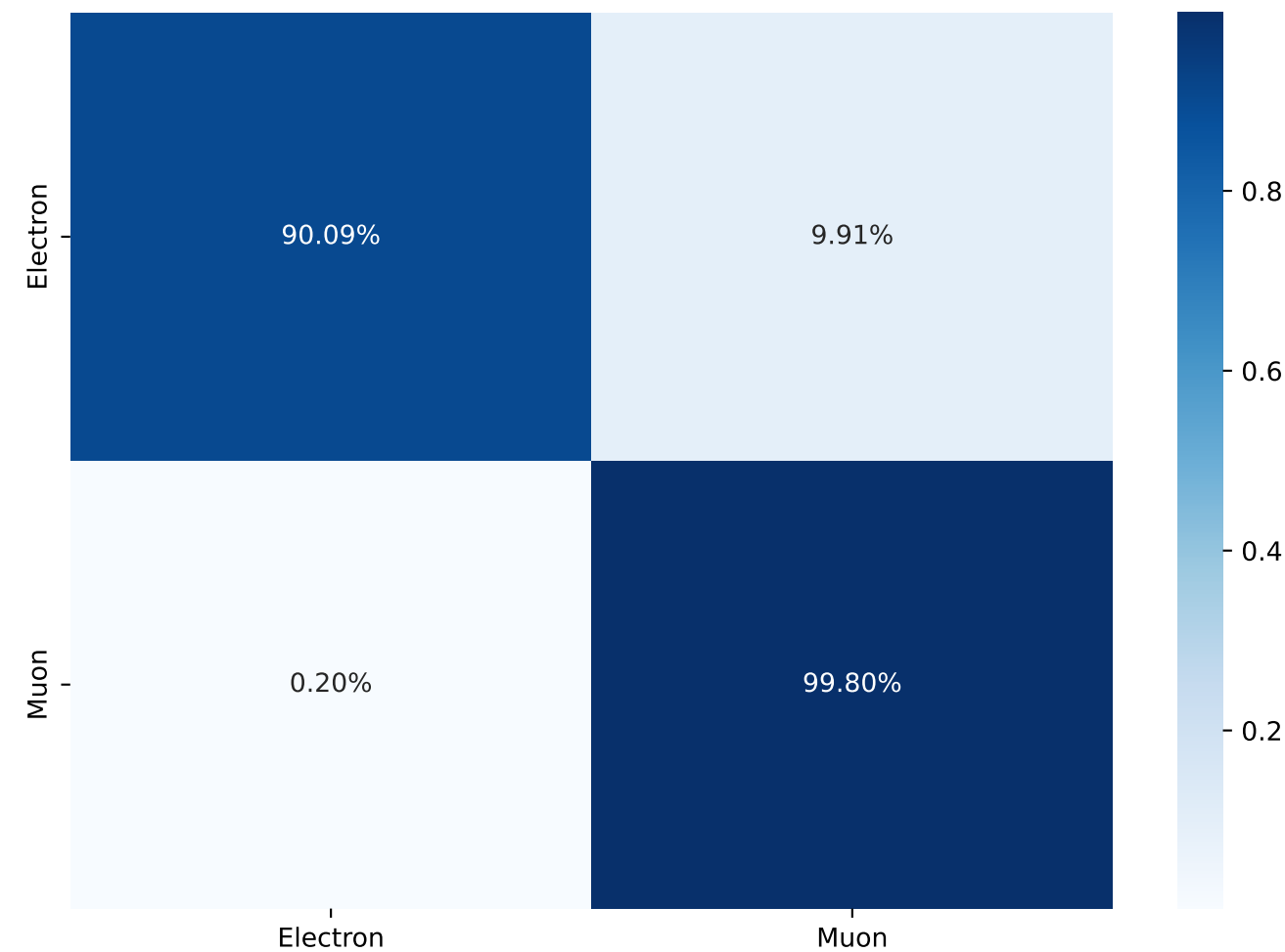


Feature Importance

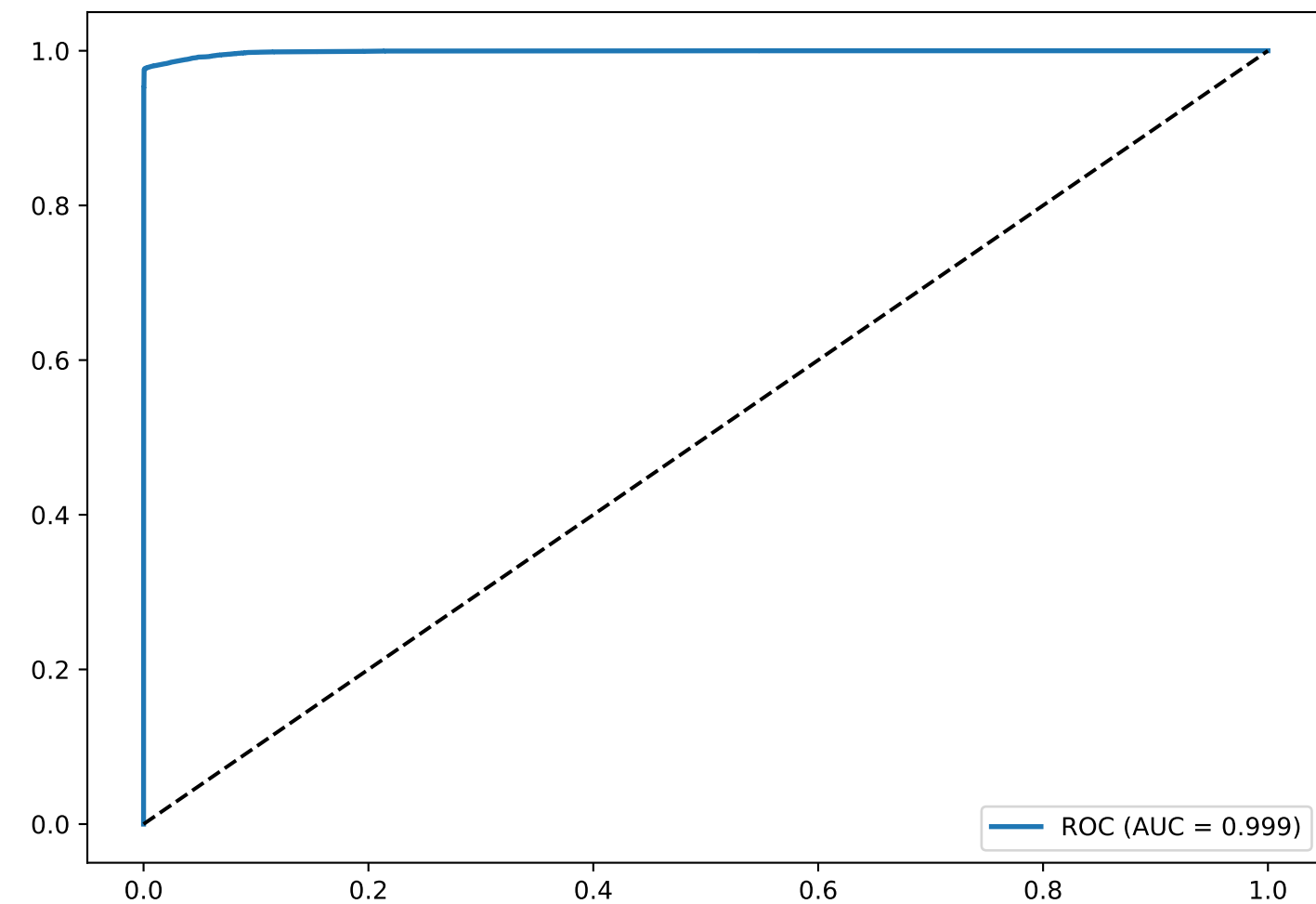


EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.981

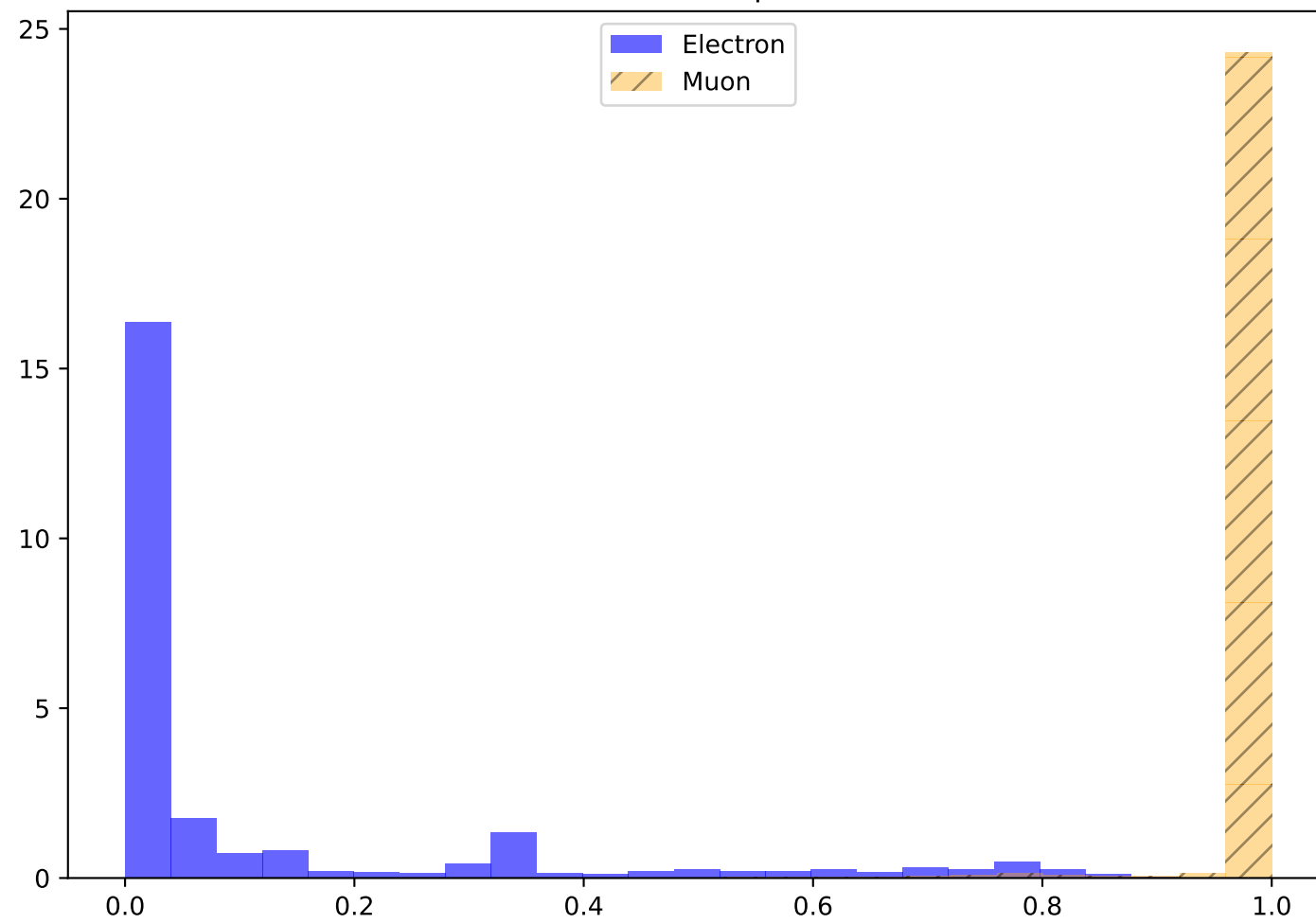
Confusion Matrix



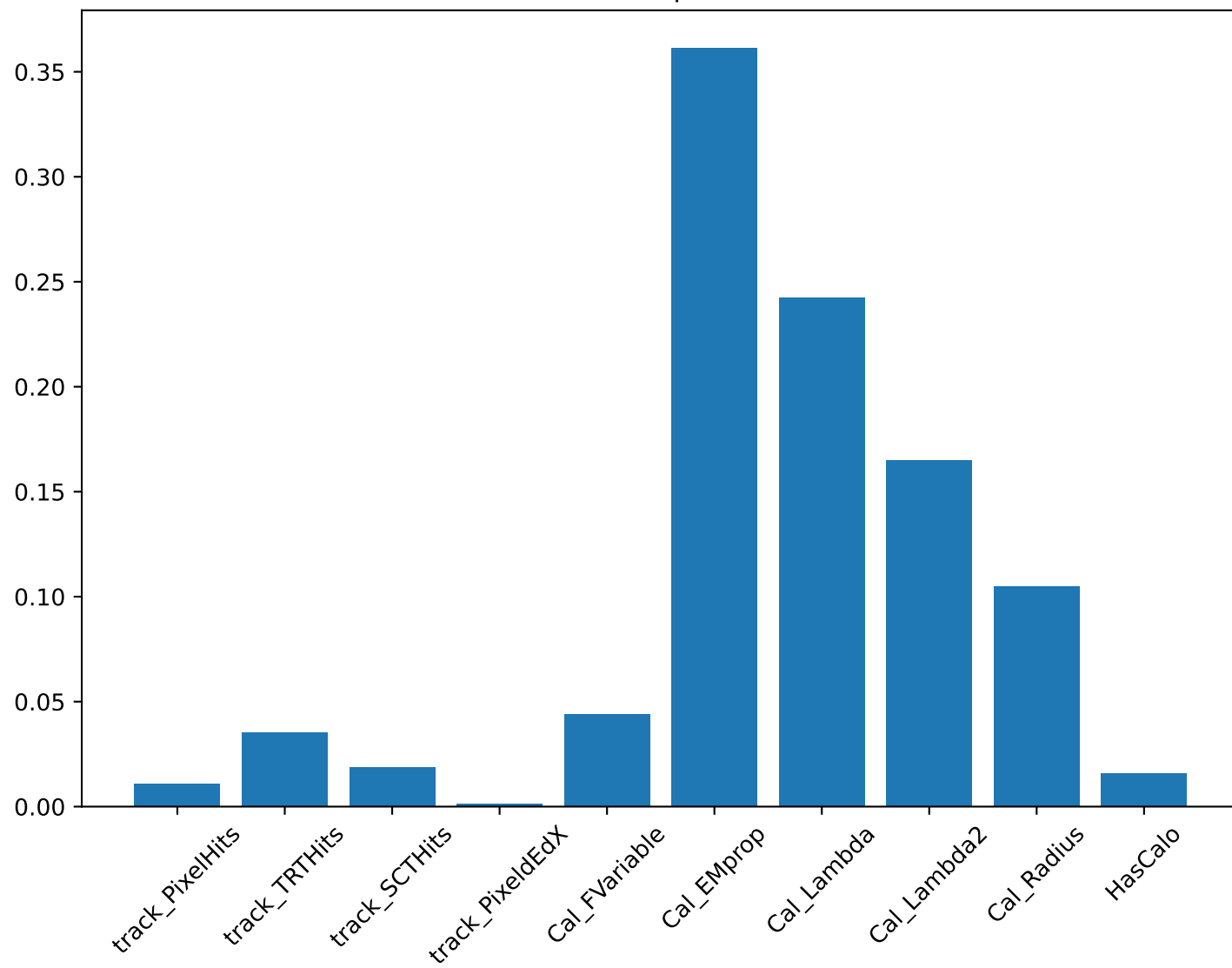
ROC Curve



Classifier Response

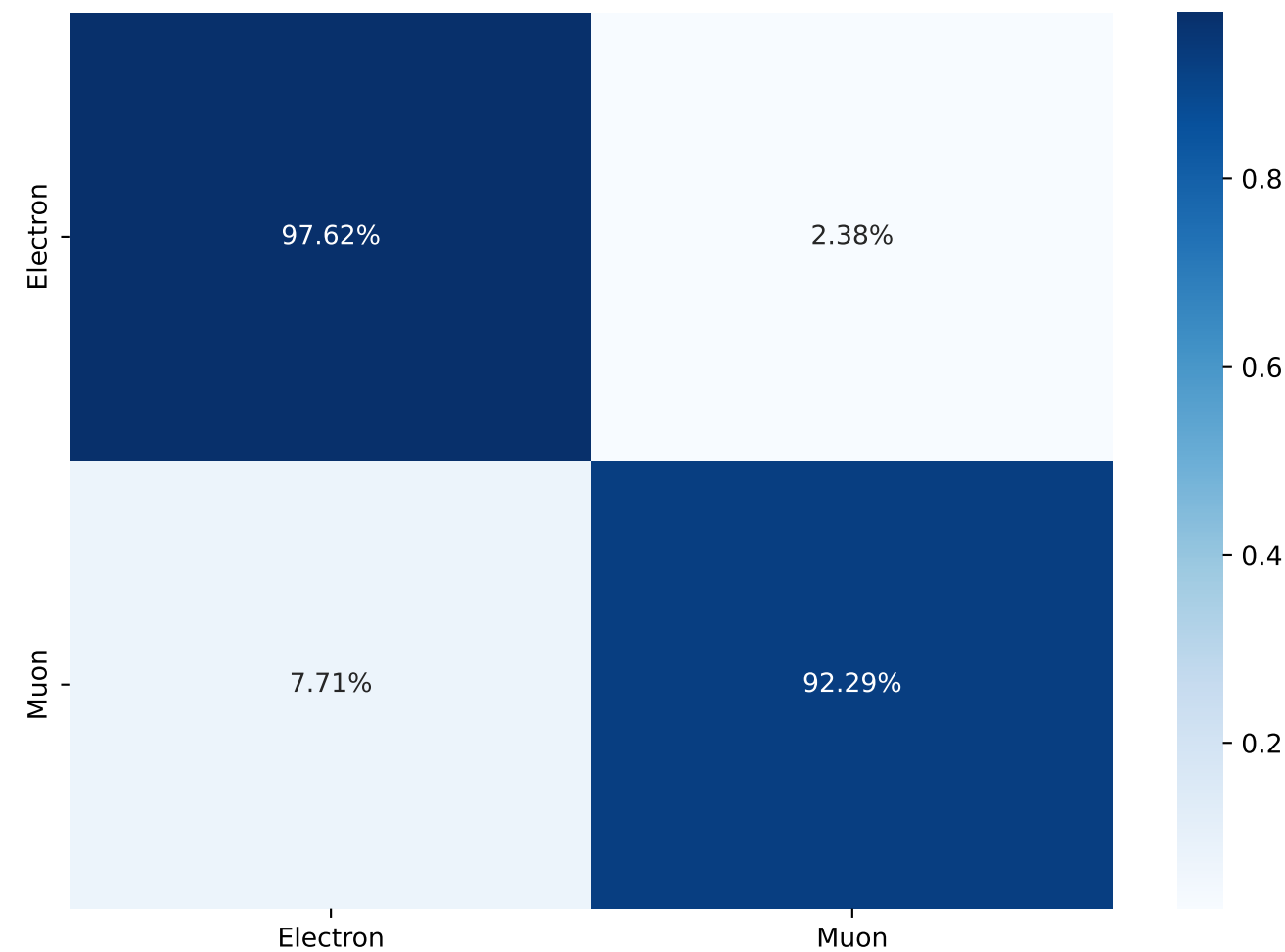


Feature Importance

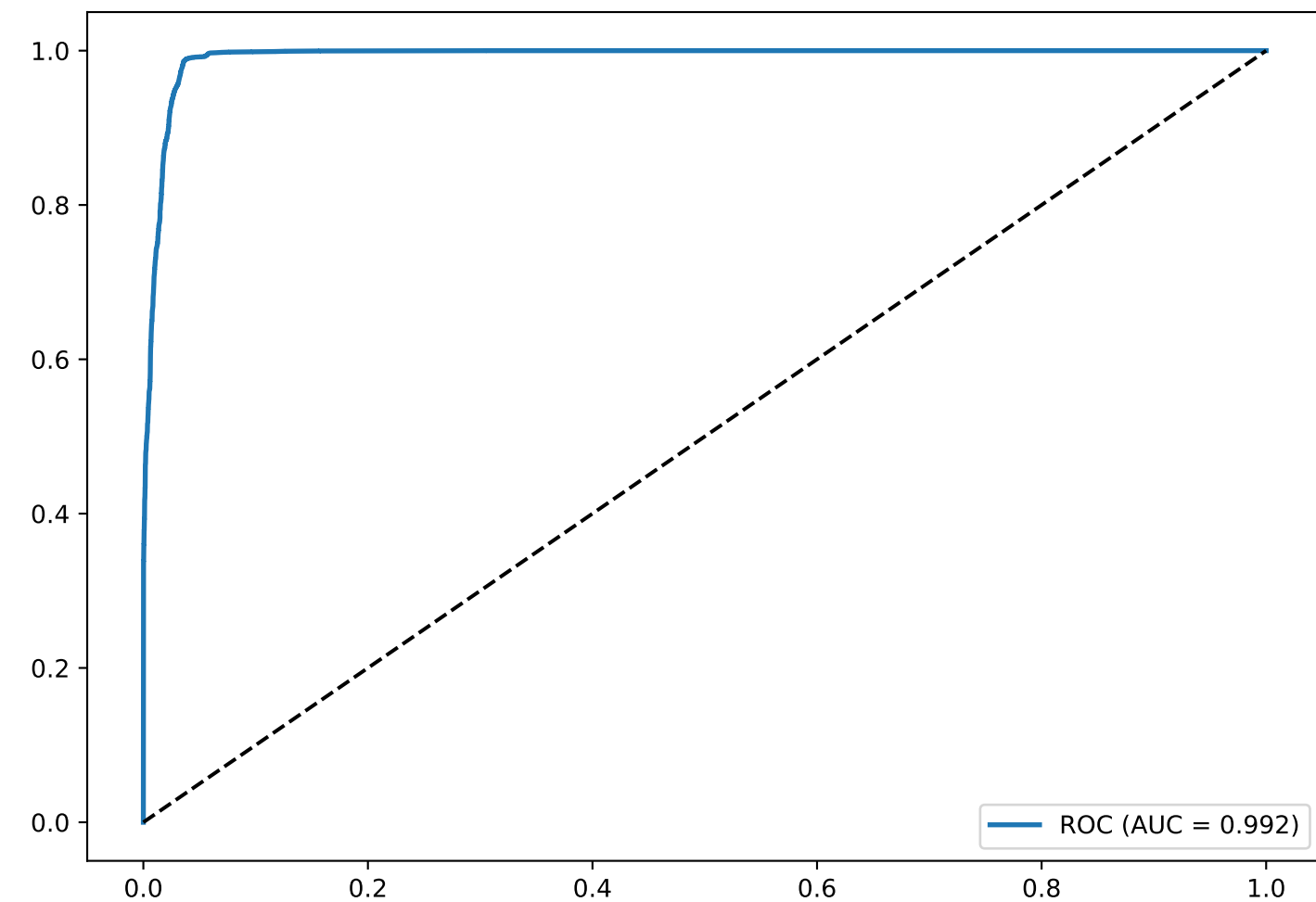


EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.929

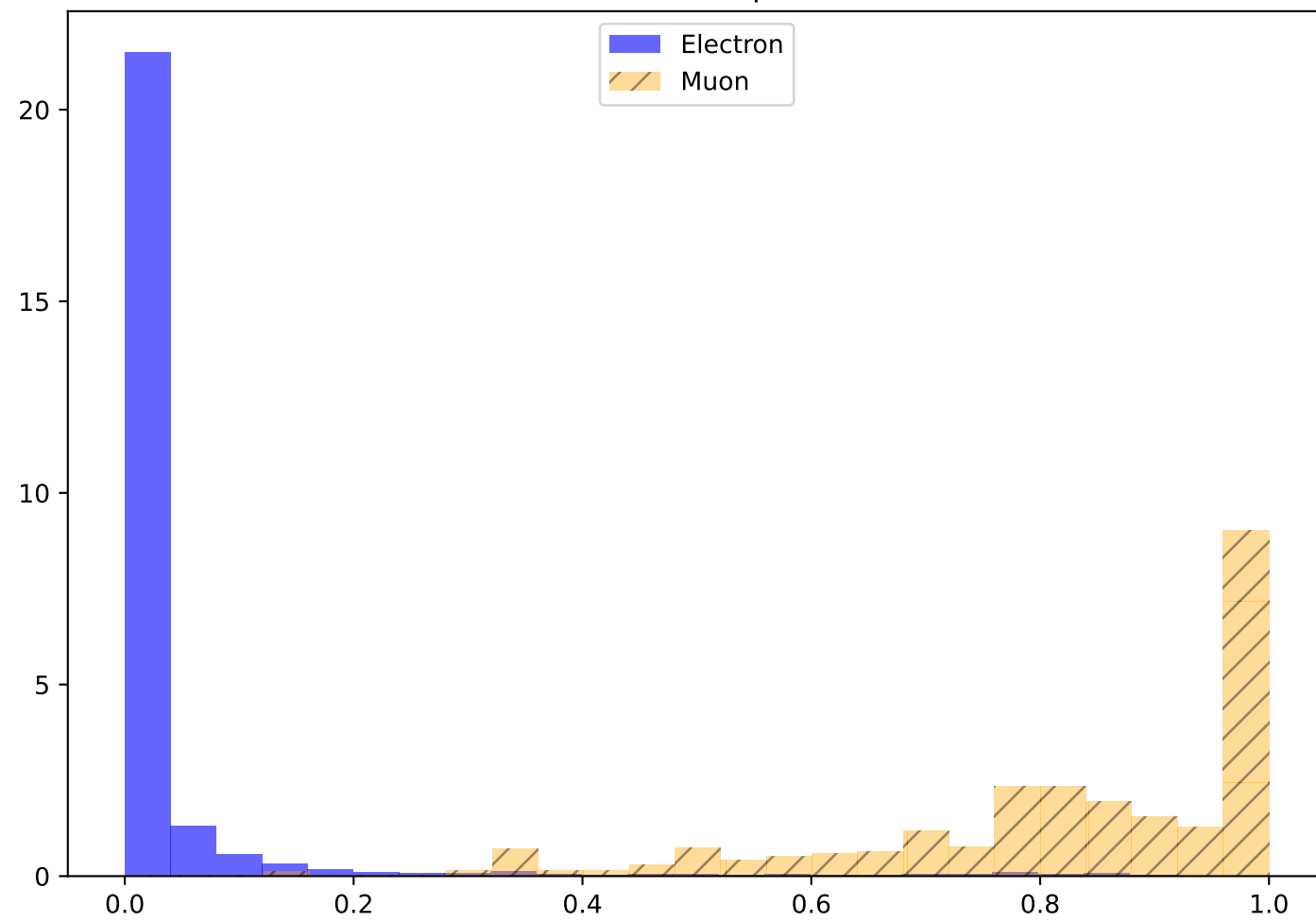
Confusion Matrix



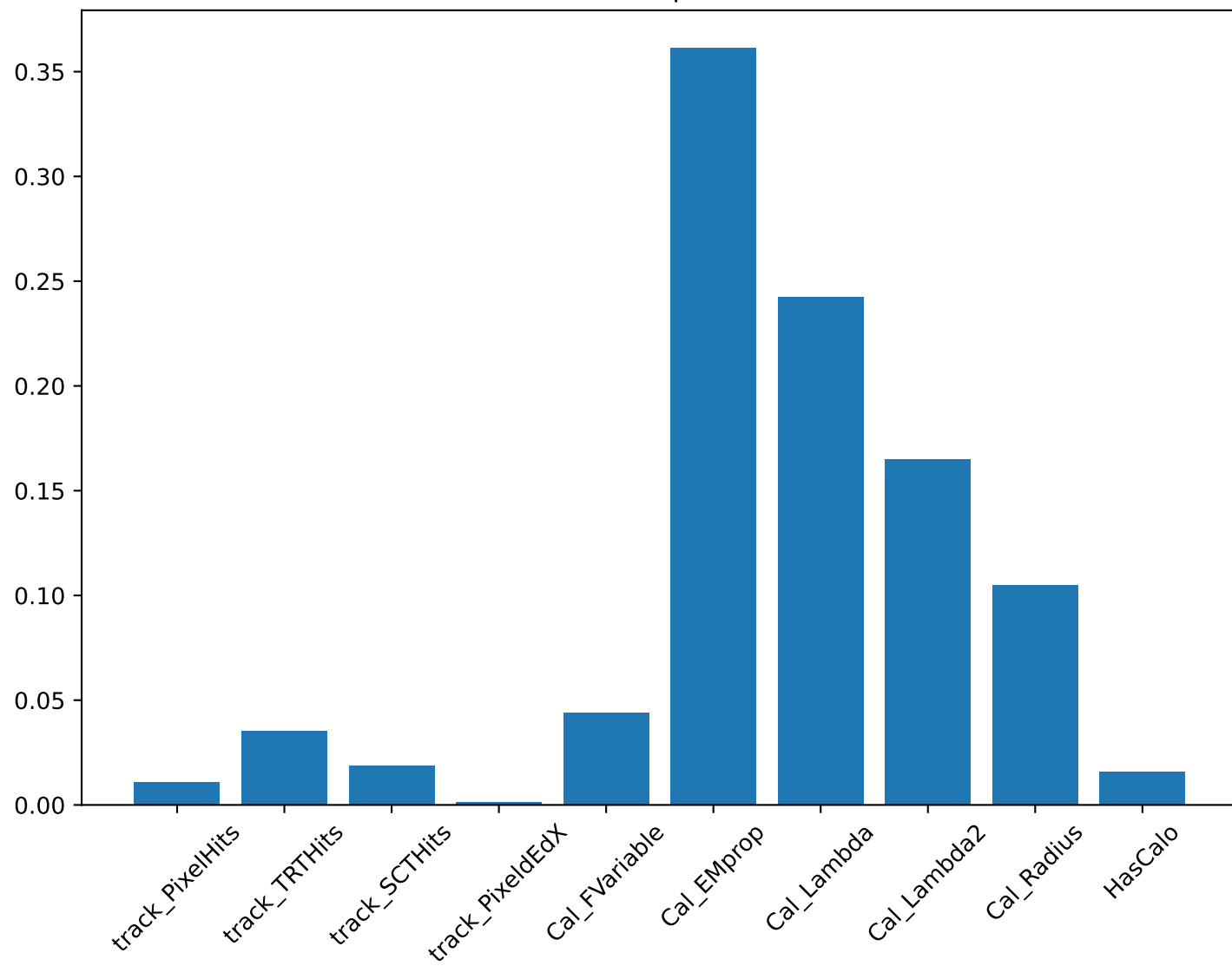
ROC Curve



Classifier Response



Feature Importance



Classification Reports Summary

Trained: Double TrueData.root | Test: TrueData.root

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	0.85	0.93	0.89	72303
Muon	0.99	0.97	0.98	352640
accuracy			0.96	424943
macro avg	0.92	0.95	0.93	424943
weighted avg	0.96	0.96	0.96	424943

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	0.91	0.94	0.92	44856
Muon	0.98	0.98	0.98	184731
accuracy			0.97	229587
macro avg	0.95	0.96	0.95	229587
weighted avg	0.97	0.97	0.97	229587

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.99	0.90	0.94	15581
Muon	0.98	1.00	0.99	73943
accuracy			0.98	89524
macro avg	0.98	0.95	0.97	89524
weighted avg	0.98	0.98	0.98	89524

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.62	0.98	0.75	11866
Muon	1.00	0.92	0.96	93966
accuracy			0.93	105832
macro avg	0.81	0.95	0.86	105832
weighted avg	0.95	0.93	0.94	105832